

A1 production kernel manifest – promotion-package reconciliation (v1.6)

TECT verification-first repository · claim A1-PRODUCTION-KERNEL-MANIFEST

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1. Purpose and scope

This note reconciles the promotion package for the canonical N-001 production-kernel manifest. It records only the pure-Brazovskii scalar slice: the full anisotropic, locked, shell-biased, condensate, and Hessian operators remain outside scope. The operator has not yet independently certified this result, so the claim remains T1.

2. Verified runtime configuration

The verifier reads the six physical fields, three tolerances, and the complete seven-key `scalar_slice` from [claims/A1-PRODUCTION-KERNEL-MANIFEST/canonical_n001_kernel.json](#). It imports the original vendored N-001 solver sources and obtains

$$\text{bloch_matrix_linear}(|k|, \{r, Z, Y, q_0, \text{**scalar_slice}\}) = r_{\text{zero}} + Z|k|^2 + Y|k|^4$$

with maximum residual 8.9×10^{-16} . The stored source hashes, the three independent consistency deltas, and the positive shell mass are all checked.

3. Reproduction and record ownership

`codes/foundations/a1_kernel_checks.py` v1.6 runs 14 self-tests and writes its result to [claims/A1-PRODUCTION-KERNEL-MANIFEST/runs/a1_kernel_checks.json](#). This keeps the numerical artefact with the claim whose manifest it supports rather than with an analytic dependency. The expected terminal line is `A1 kernel checks v1.6: 14/14 PASS`.

4. Packaging decision

The existing v1.5 directory is pre-certification reproduction evidence. It is not a final T5 package: its note states that certification is pending and its manifest does not carry a stamped source commit. If the operator certifies the result and designates it as main-line, a new claim-top-level bundle must be built last with the confirmed T5 note, a tier-and-date-stamped name, a source commit, and final release checks. The withdrawn Math424 certificate is not evidence for this package.

5. Devil’s-advocate

(α) “*The checker chooses the scalar slice.*” DISMISSED — every setting is loaded from JSON and demonstrated load-bearing. (β) “*The vendored solver is not original.*” DISMISSED — all three source hashes are checked against provenance and config. (γ) “*Scalar-slice agreement proves the full operator.*” VALID with mitigation — the scope excludes the full operator. (δ) “*A PASS is certification.*” UPHELD — independent operator reproduction is still required, so no tier action is taken here.

6. Result footer

Result ID:	A1-PRODUCTION-KERNEL-MANIFEST
Precise statement:	The canonical N-001 pure scalar slice passes the stored-field, runtime-symbol, positivity, and source-hash checks (14/14).
Scope:	Pure-Brazovskii scalar slice only; full operator excluded.
Dependencies:	A1-KERNEL-IDENTITY; A1-SCALAR-ANALYTIC-BRANCH.
Evidence grade:	ANALYTIC + EXECUTED.
Reproduction command:	python codes/foundations/a1_kernel_checks.py
Expected output:	A1 kernel checks v1.6: 14/14 PASS.
Falsification gate:	Config, scalar-slice, source-hash, delta, or coefficient mismatch.
Tier before / after:	T1 / T1 (promotion package reconciled; no certification).
No-overclaim statement:	Not operator-certified; no full-operator or theorem claim.
Next required action:	Independent operator reproduction, main-line decision, then T5 packaging.